

Adult (Census Income) — Day 1 Analysis Summary

NetixSol Internship | Prediction of income >50K/year | 1-page deliverable

1. Problem Framing

Binary classification task: predict whether an individual earns **>50K/year** using demographic and job features (age, education, occupation, hours/week, etc.). The dataset has **48,842 samples** and 15 features. The positive class (>50K) represents the high-earner population we aim to reliably target.

2. Chosen Metric & Business Objective

Business objective: Maximize precision among contacted users to avoid wasted outreach.

Class base rate: 23.93% of the dataset earns >50K. Because the classes are imbalanced, accuracy is misleading — a naive 'always predict <=50K' model reaches 76% accuracy while catching zero high earners. The primary metric is therefore **Precision**: the fraction of people we predict to earn >50K who actually do. High precision means less budget wasted contacting non-qualifying individuals.

3. Baseline Results (Hold-Out Test Set)

Baseline	Acc	Prec	Rec	F1	ROC AUC	PR AUC
Majority (always <=50K)	0.7607	0.0000	0.0000	0.0000	0.5000	0.2393
Rule (education-num >= 13)	0.7530	0.4844	0.4970	0.4906	0.6653	0.3611

The **rule-based baseline wins**: it catches half the high earners (recall 49.7%) with the majority classifier catching none. However, only 48.4% of its predicted high earners actually qualify. A real model must beat the rule baseline's F1 of 0.49 — targeting F1 >= 0.60 with precision >= 65% — to justify real-world outreach cost.

4. Error Analysis & Next Steps

Sampling **1,237 false positives** and **1,176 false negatives** from the rule baseline revealed:

- False positives: Bachelors+ holders in low-paying sectors (Private/Unknown), low capital-gain, often job-matched work hours below the high-pay threshold.
- False negatives: No degree but 45-60 hrs/week, some with capital-gain > 0, older workers (40-60) who built income via experience.

Issues to fix: log-transform/bin skewed capital-gain & capital-loss; group native-country into US vs Other; drop redundant education column; group marital-status into Married vs Not-Married; group rare occupations; cap hours-per-week outliers.

Primary metric for the rest of the week: Precision. Backed by a baseline of 48.4% precision (49.7% recall, F1 0.49), the goal is to raise precision above 65% while holding recall above 40% — a concrete, business-relevant improvement over the rule baseline.